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Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Digitales Design (DiD)

Speicherelemente und FlipFlops

LAT

Studiengang Systemtechnik
Studiengang Energie und Umwelttechnik
Studiengang Informatik und Kommunikationssysteme

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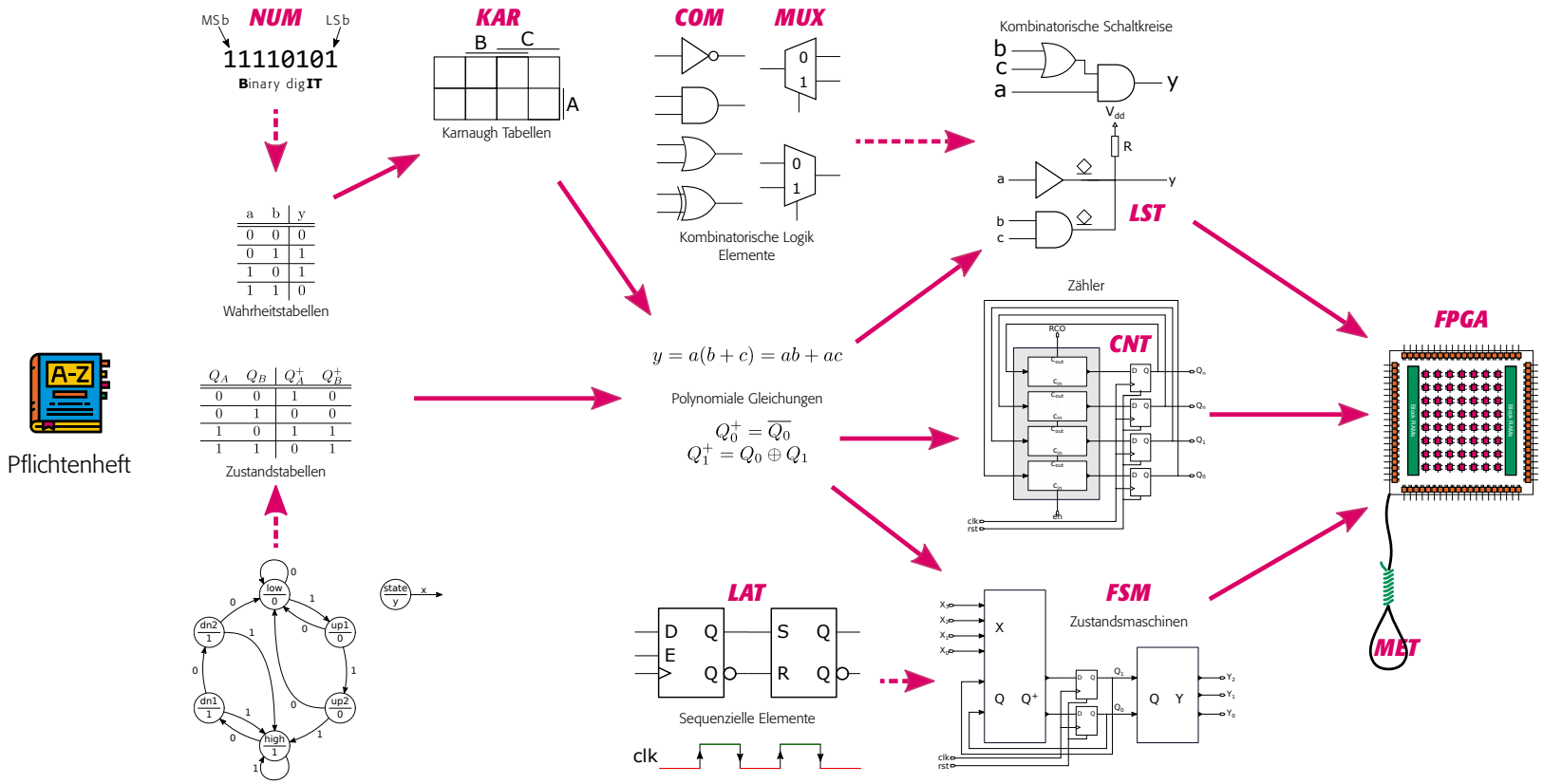
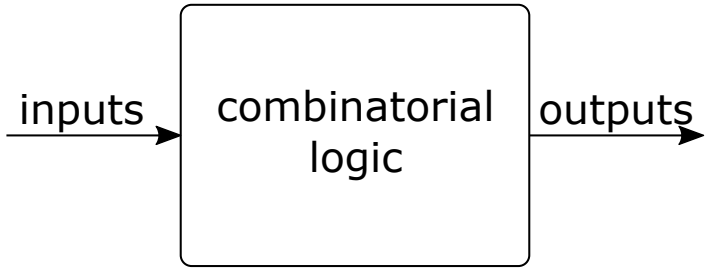
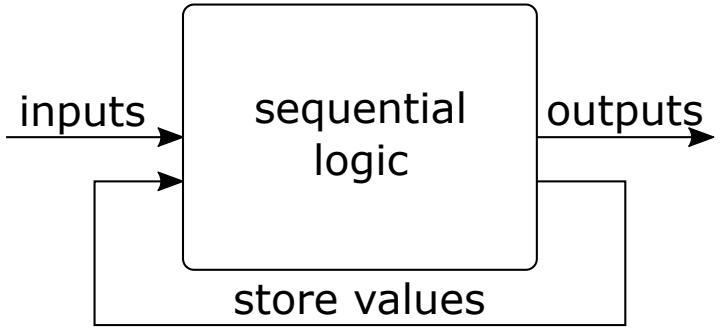
Christophe Bianchi christophe.bianchi@hevs.ch
François Corthay francois.corthay@hevs.ch

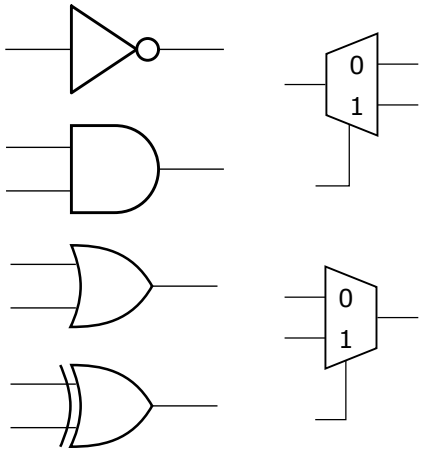
Aktueller Inhalt des Themas im Kurs

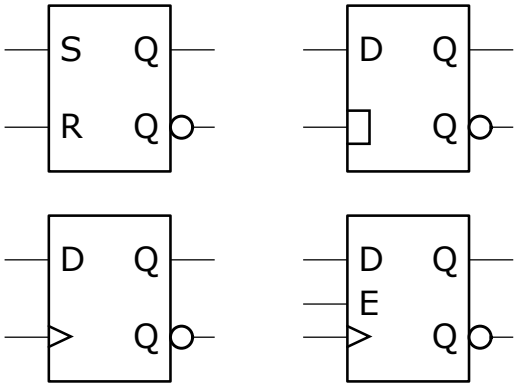


Kombinatorische und sequentielle Logik









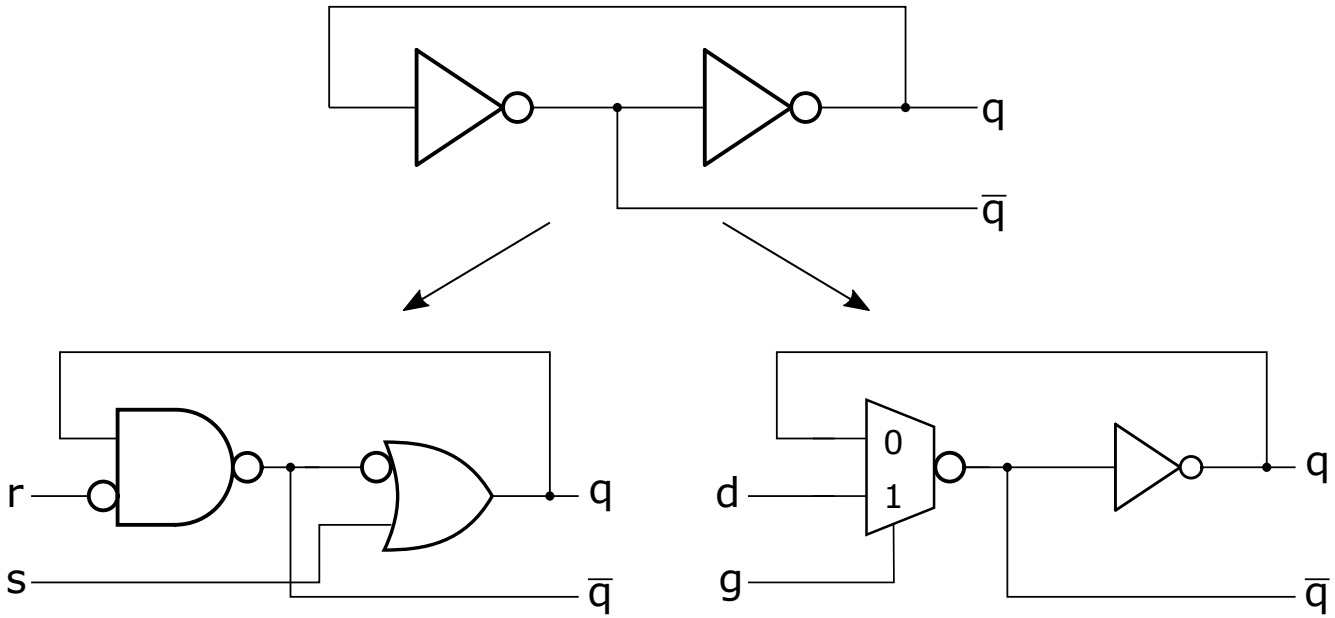
Inhalt

- **Speicherelemente (Latch)**
  - SR-Speicherelement (SR-Latch)
  - Charakteristische Gleichung
  - D-Speicherelement (D-Latch)

- FlipFlops

SR-Speicherelement

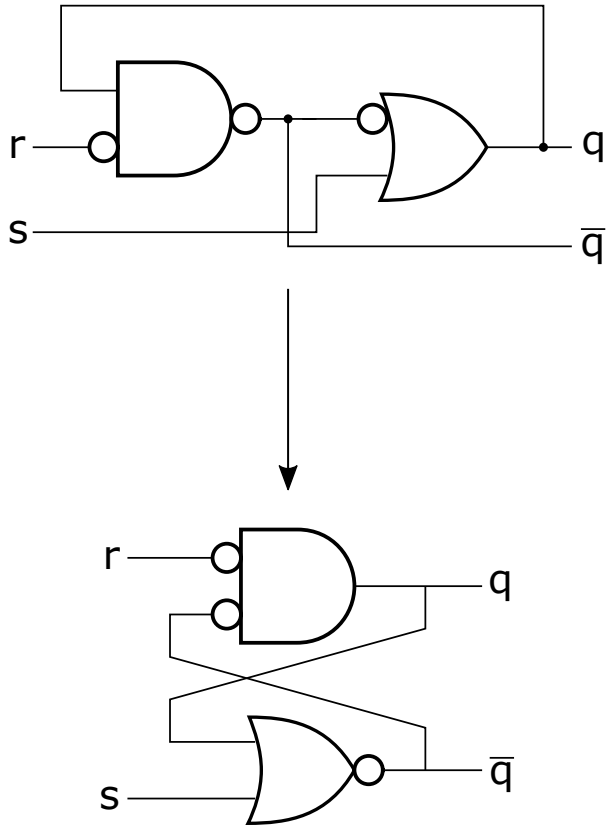
SR-Latch

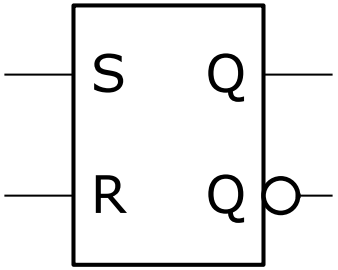


2 gekoppelte Inverter bilden ein Speicherelement

SR-Speicherelement

SR-Latch (set-reset)



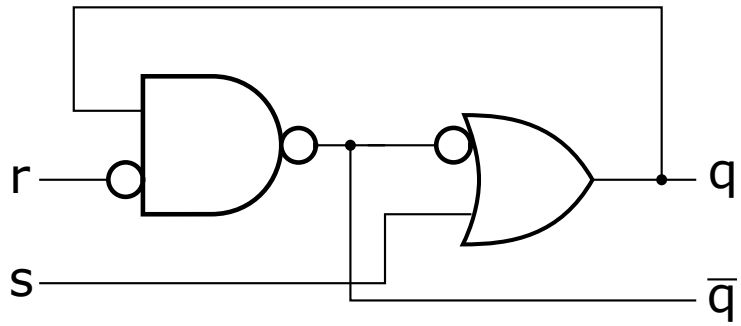


s	r	q	qn	Funktion
0	0	Unverändert		Speicherung
0	1	0	1	Nullsetzung (reset)
1	0	1	0	Setzen auf 1 (set)
1	1	0	0	Verboten

SR-Speicherelement

SR-Latch charakteristische Gleichung

Die charakteristische Gleichung beschreibt die Funktionalität des Speicherelements



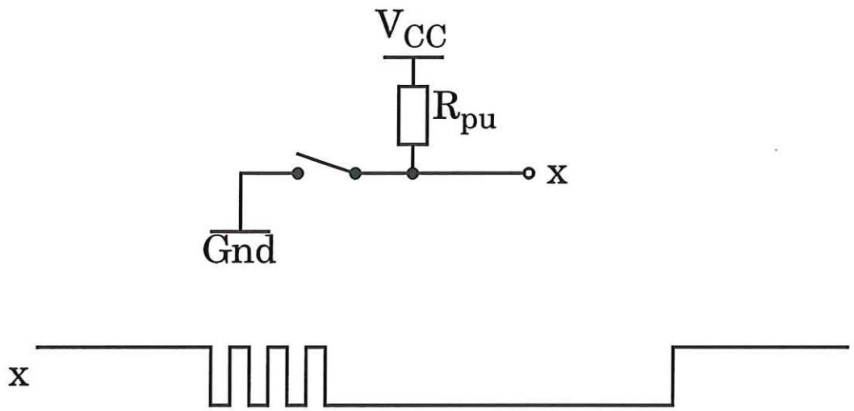
$$q = s + \overline{r}q$$

« q » auf beiden Seiten der Gleichung zeigt die Speicherschleife (q=q bei s=0 und r=0)

Aufgabe 2.1

Anti-Prell-Schaltung

Mit Hilfe von einem Umschalter und einem Speicherelement, entwerfen Sie eine Schaltung, welche ein prellfreies Signal liefert.



Aufgabe 2.1

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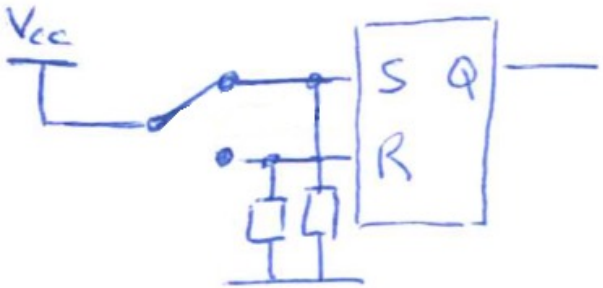
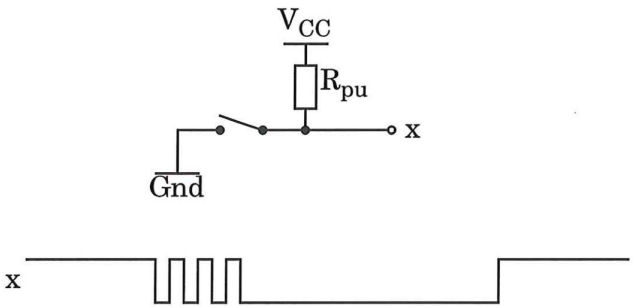
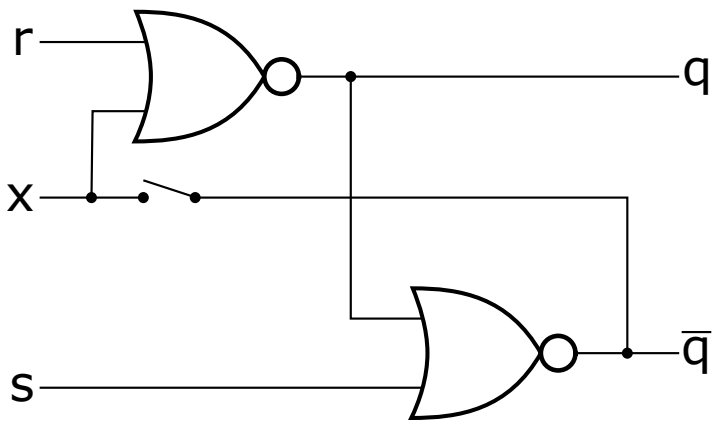




SR-Speicherelement

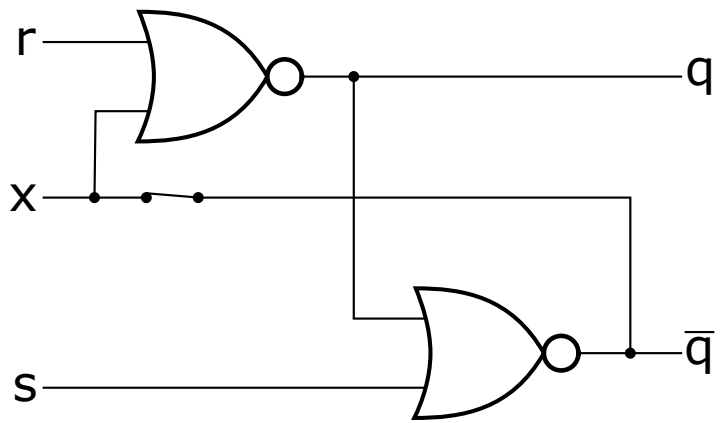
Funktionsanalyse (Kombinatorisches Model)



s	r	x	q	qn	
0	0	0	1	0	
0	0	1	0	1	
0	1	0	0	1	
0	1	1	0	1	
1	0	0	1	0	
1	0	1	0	0	
1	1	0	0	0	
1	1	1	0	0	

SR-Speicherelement

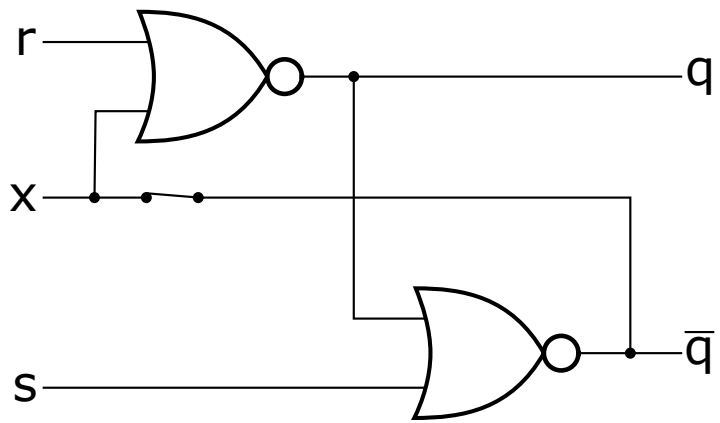
Funktionsanalyse (Kombinatorisches Model)



s	r	x	q	qn	Gültig
0	0	0	1	0	✓
0	0	1	0	1	✓
0	1	0	0	1	×
0	1	1	0	1	✓
1	0	0	1	0	✓
1	0	1	0	0	×
1	1	0	0	0	✓
1	1	1	0	0	×

SR-Speicherelement

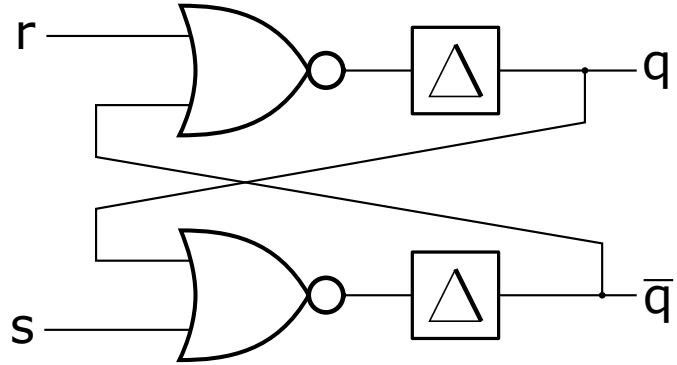
Funktionsanalyse (Kombinatorisches Model)

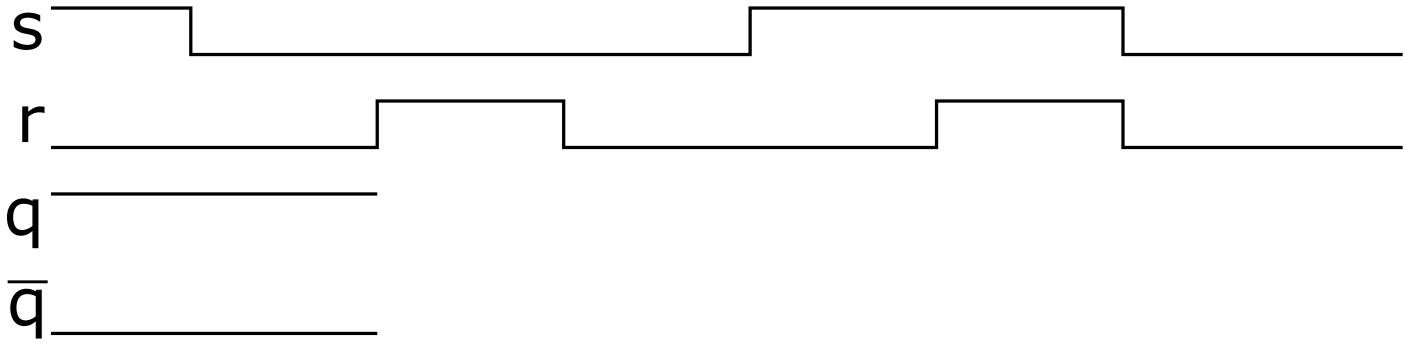


s	r		q	qn	Funktion
0	0		1	0	Speicherung
0	0		0	1	
0	1		0	1	reset
1	0		1	0	set
1	1		0	0	Verboten

SR-Speicherelement

Funktionsanalyse (Zeitliches Verhalten)





SR-Speicherelement

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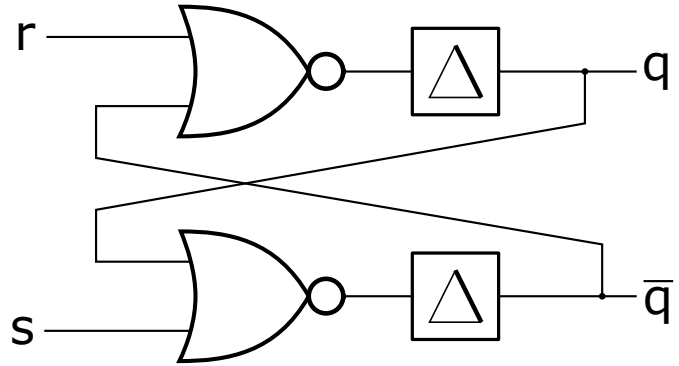

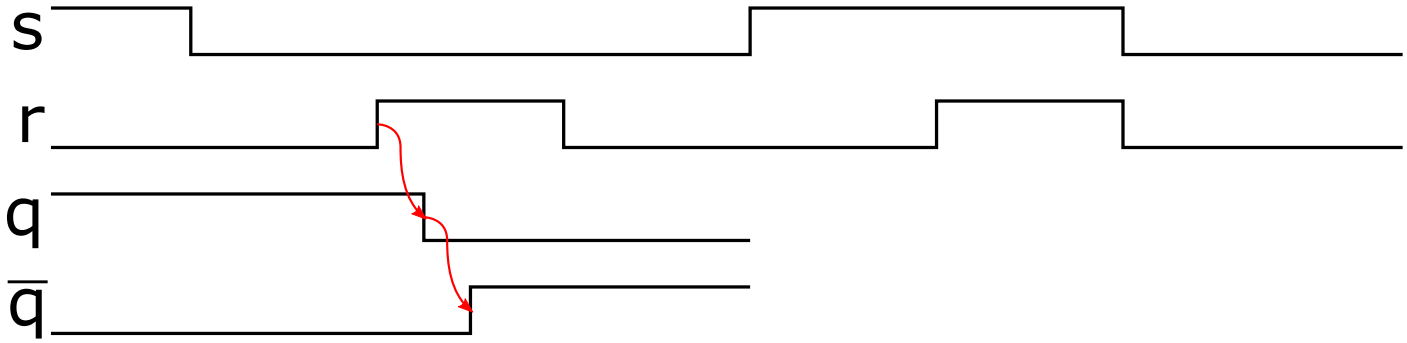

SR-Speicherelement

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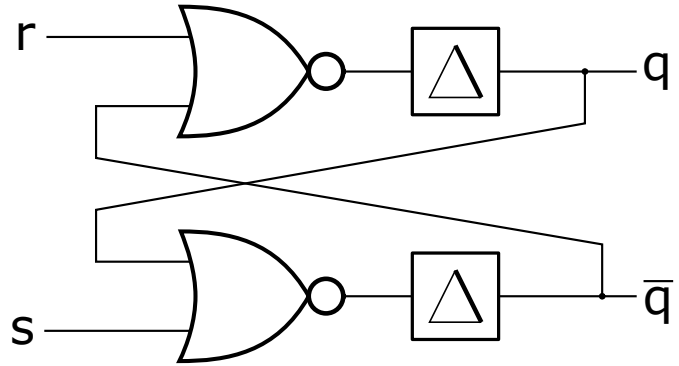
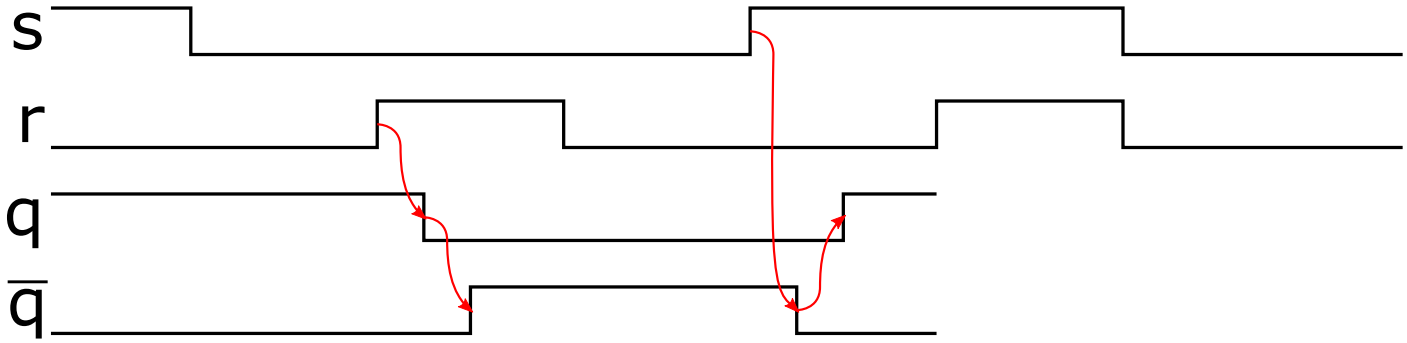
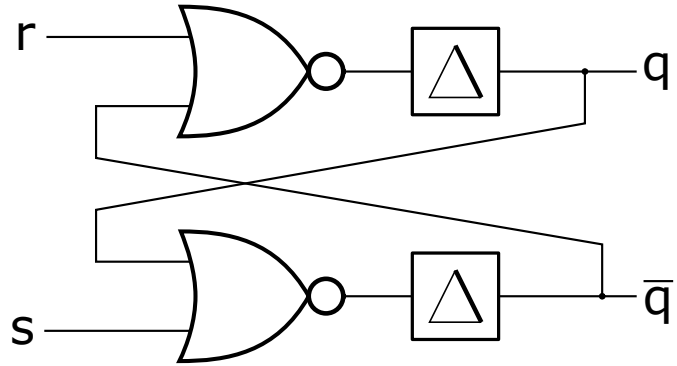
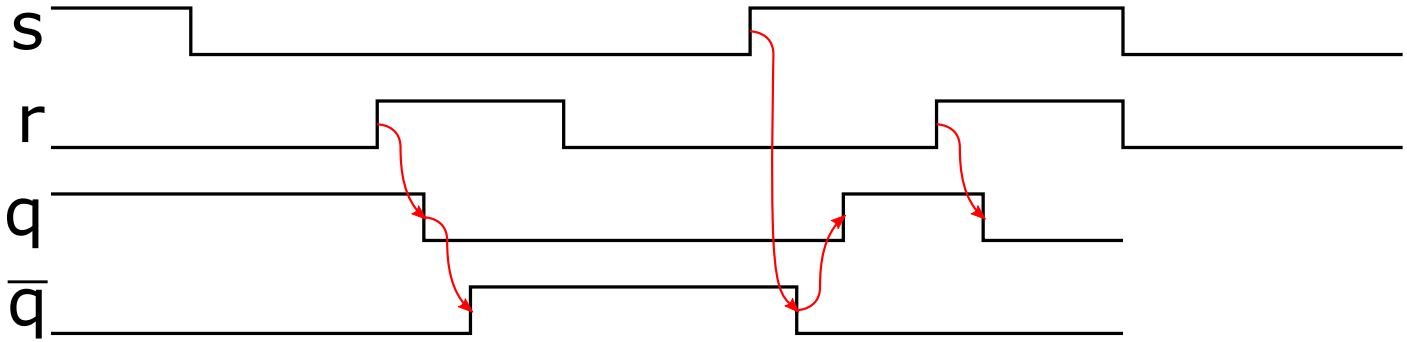




SR-Speicherelement

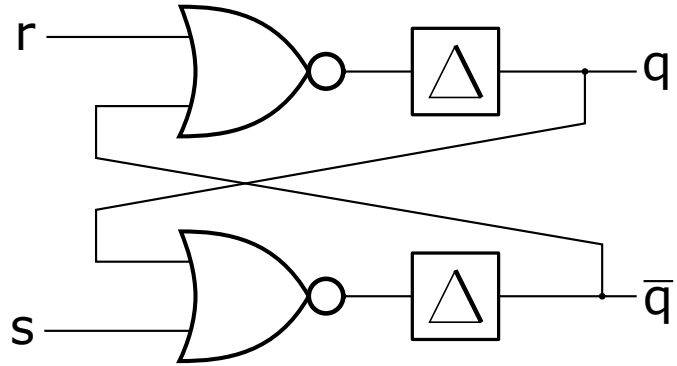
Funktionsanalyse (Zeitliches Verhalten)

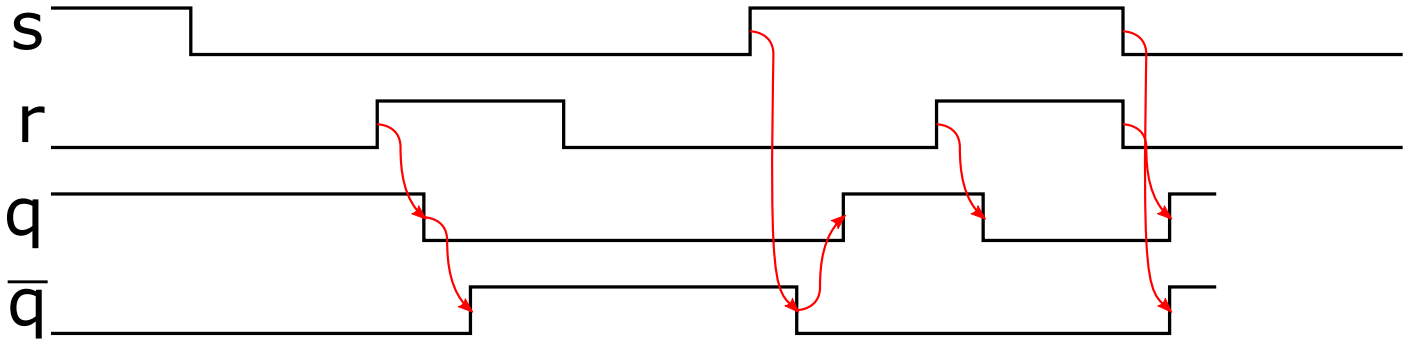




SR-Speicherelement

Funktionsanalyse (Zeitliches Verhalten)





SR-Speicherelement

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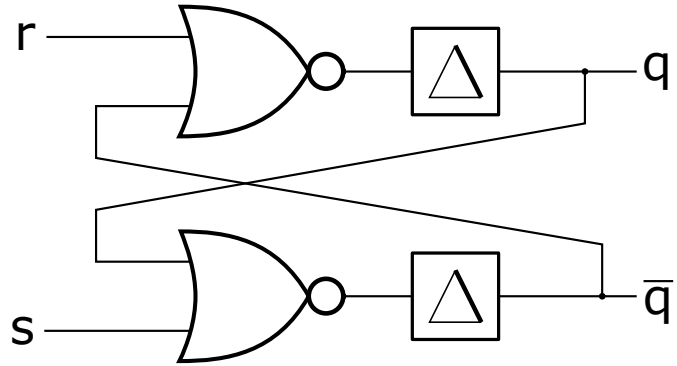
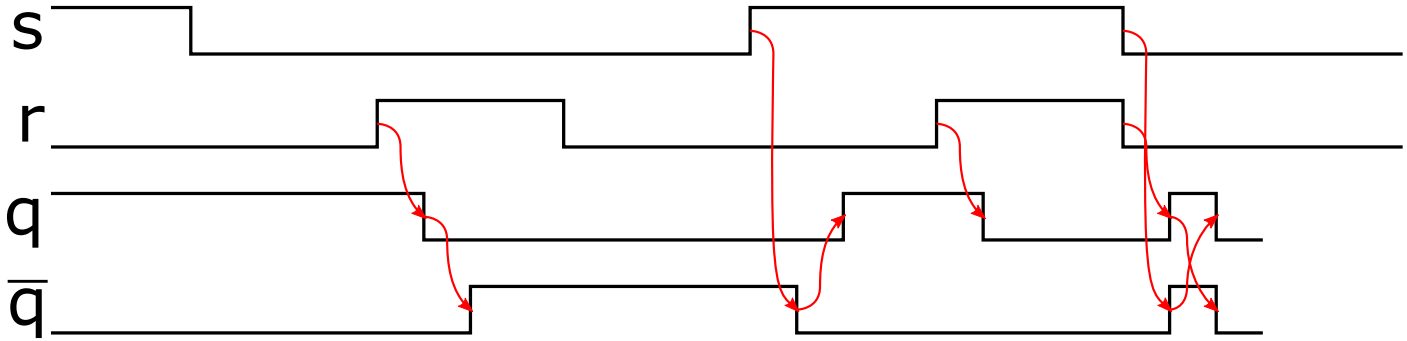
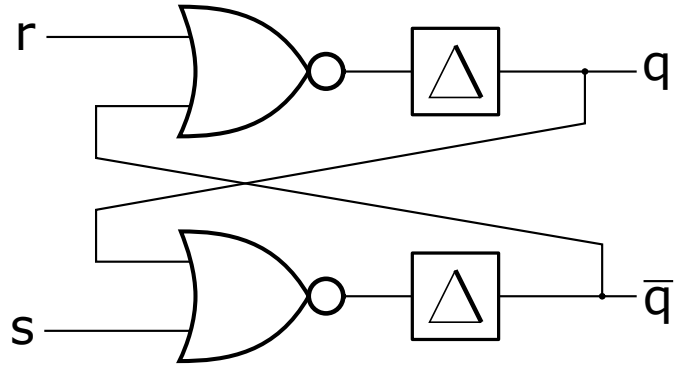
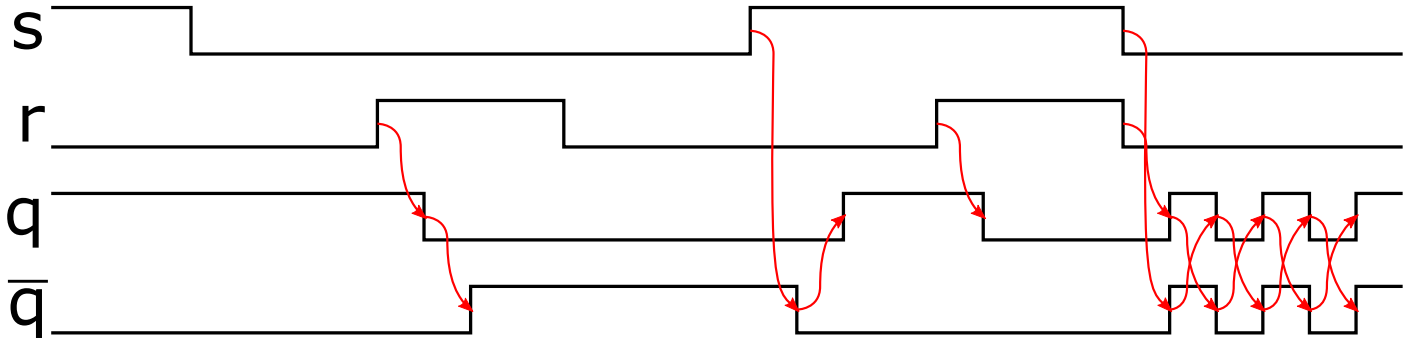




SR-Speicherelement

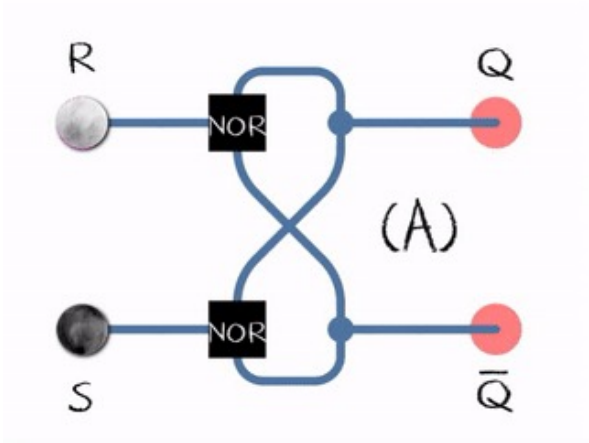
Funktionsanalyse (Zeitliches Verhalten)

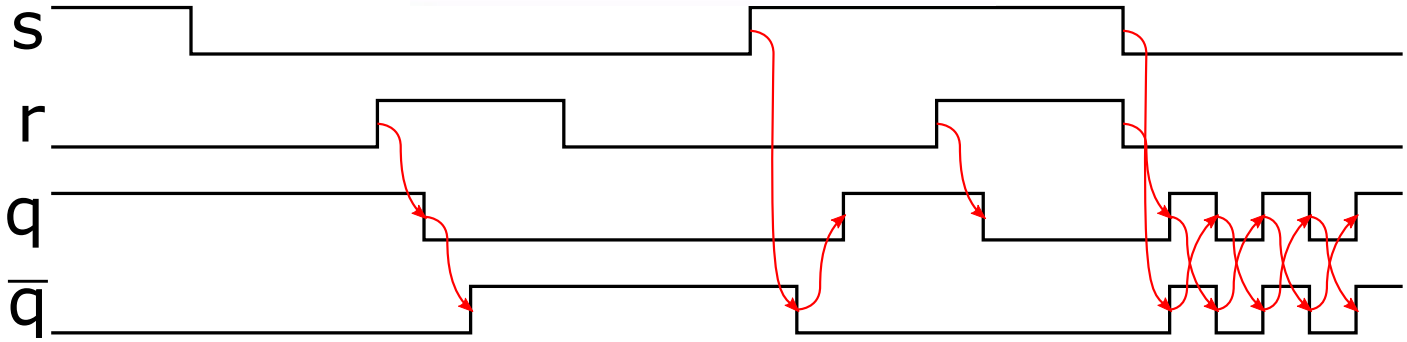




SR-Speicherelement

Funktionsanalyse (Zeitliches Verhalten)





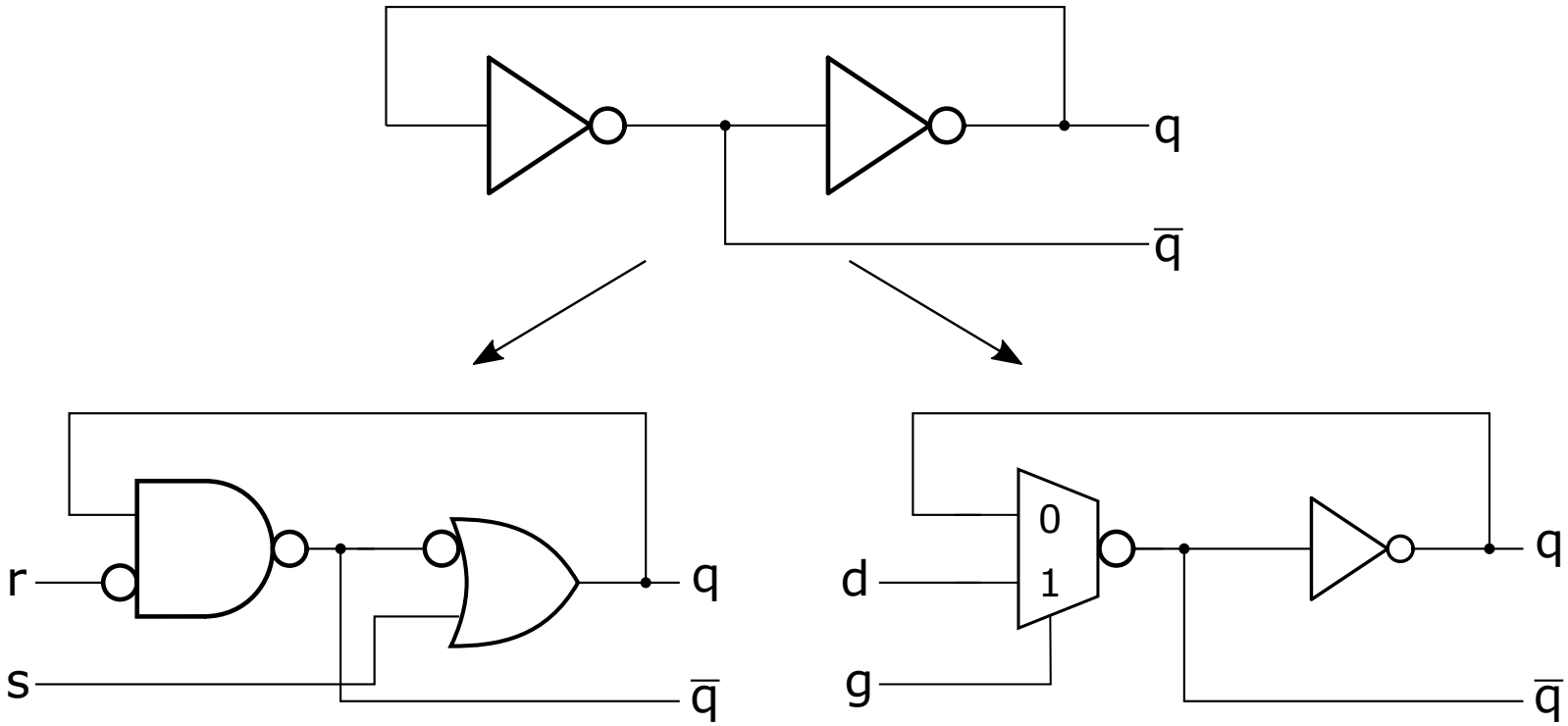
Aufgabe 2.1

Tastenauswahl

Die Tastatur eines Rechners hat 5 Tasten, um eine von 8 Operationen auszuwählen. Die Multiplikation wird gewählt, indem man auf die entsprechende Taste drückt. Die zweier Potenz wird gewählt, indem man auf die “2nd”- und danach auf die Multiplikationstaste drückt.

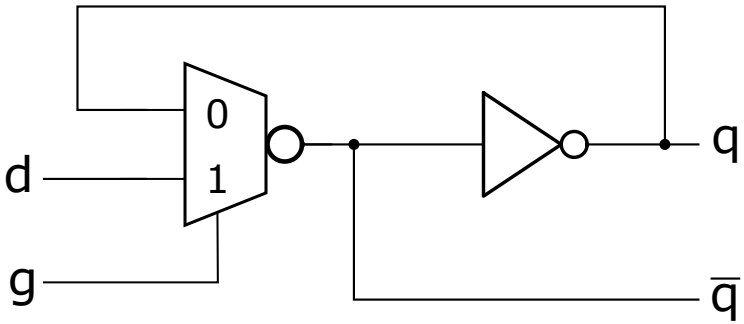
Erstellen Sie eine Schaltung mit 8 Ausgängen, welche einen Puls auf dem Ausgang liefert, welcher der gewünschter Funktion entspricht.

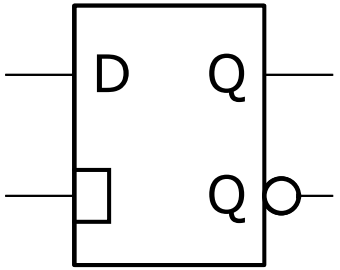
Speicherelemente



Speicherelemente

D-Speicherelement (D-Latch)



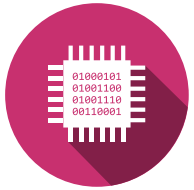


g	d	q	qn	Funktion
0	0	Unverändert		Speicherung
0	1			
1	0	0	1	Laden von D
1	1	1	0	

$$q = gd + \overline{g}q$$

Aufgabe 2.5

Synchronisation



Ein getaktetes System liefert ein Ausgangssignal, welches nur dann variiert, wenn das Taktsignal auf '0' steht.

Entwerfen Sie eine Schaltung, welche dieses Signal verspätet, bis das Taktsignal auf '1' kommt.

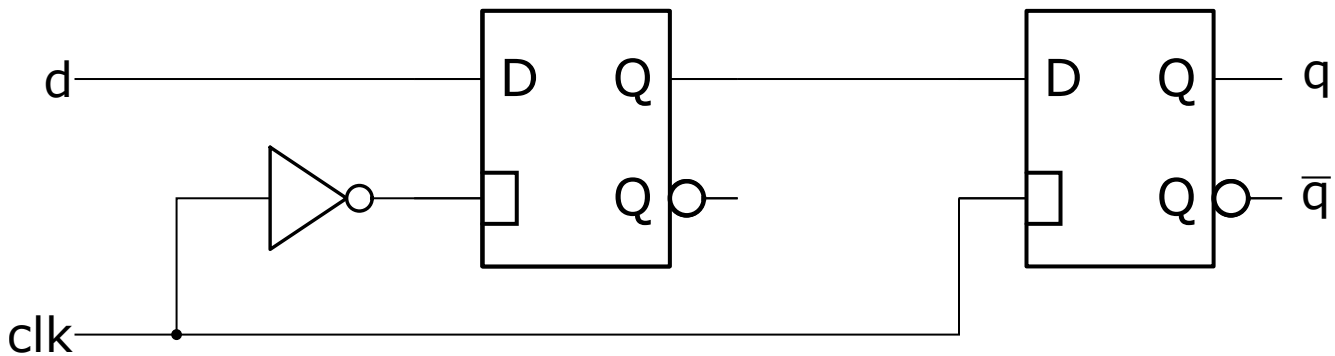
Inhalt

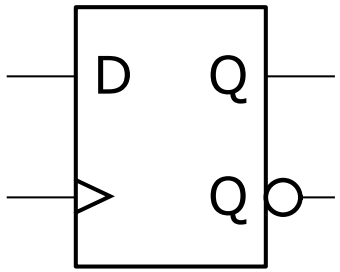
- Speicherelemente (Latch)
- **FlipFlops**
  - D-FlipFlop
  - Charakteristische Gleichung
  - SR-FlipFlop
  - E-FlipFlop
  - T-FlipFlop
  - Asynchrone Eingänge

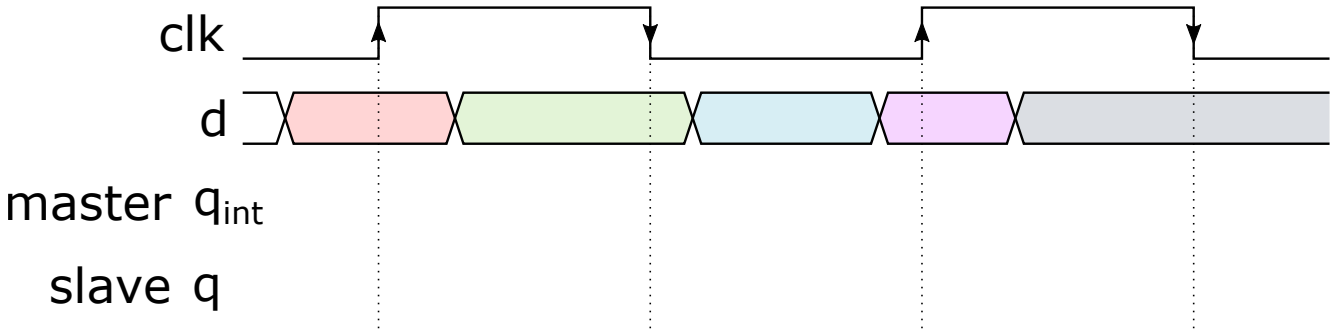
FlipFlops

D-FlipFlop

Master-Slave Latch



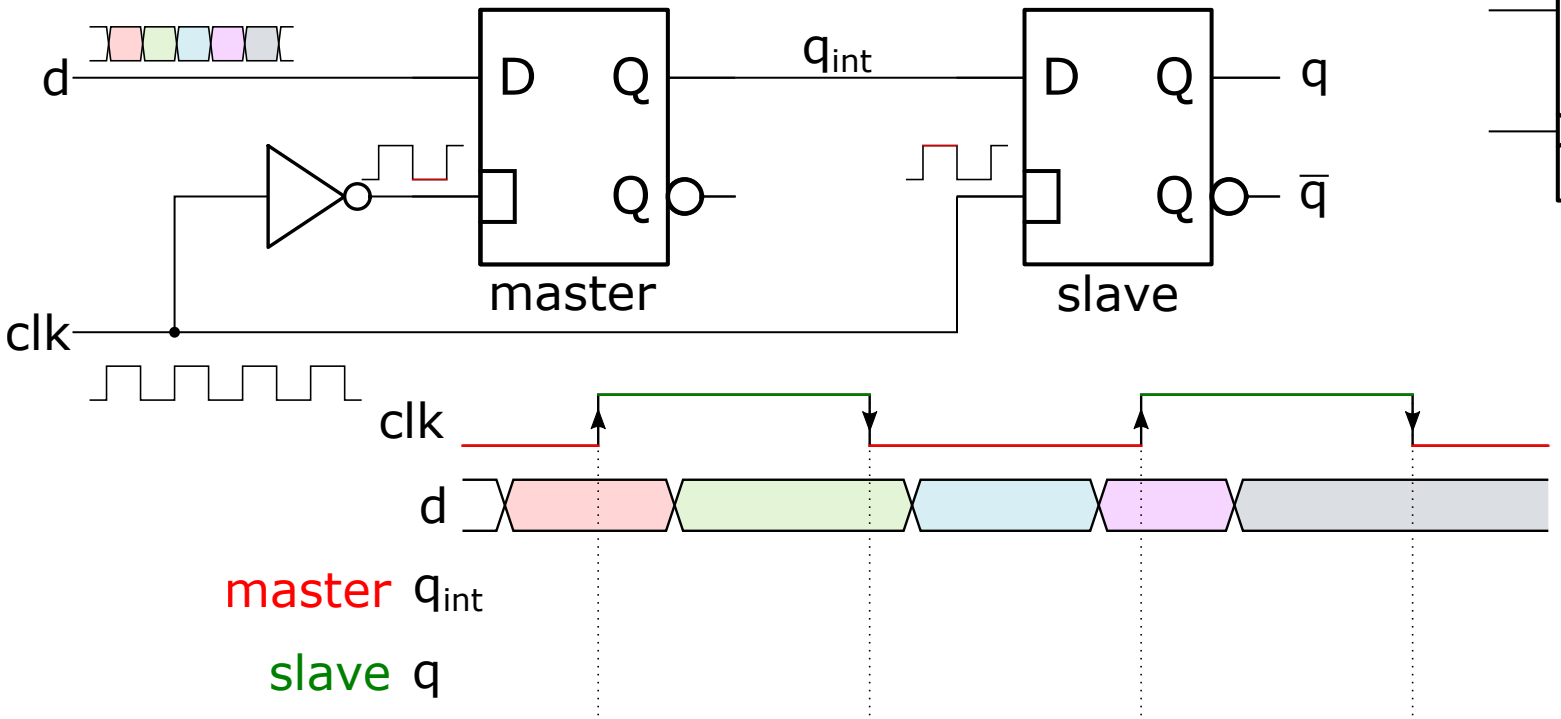


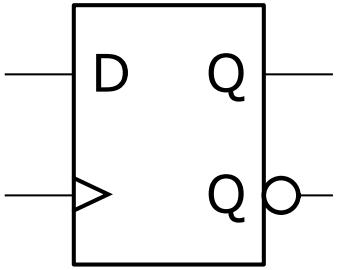


FlipFlops

D-FlipFlop

Master-Slave Latch

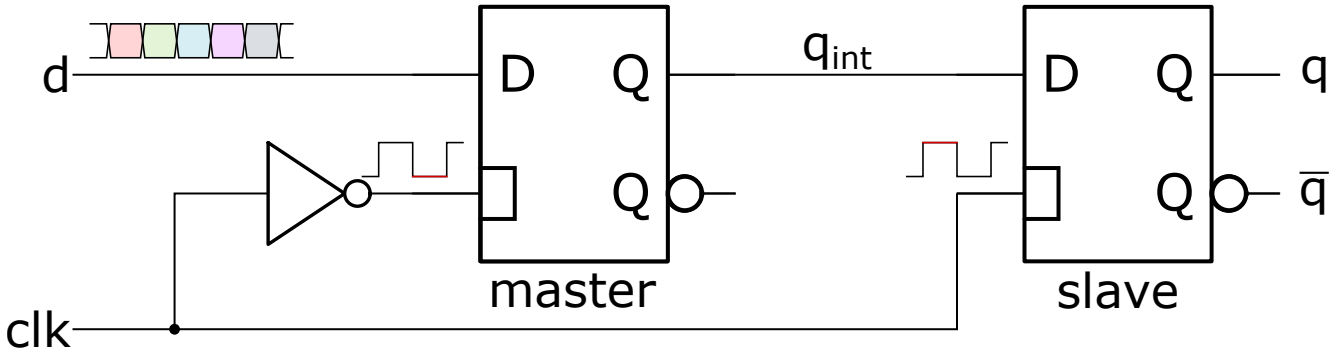


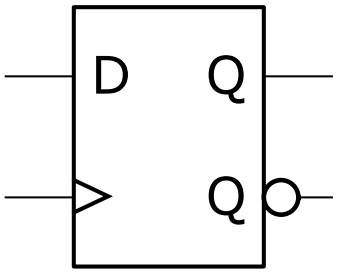


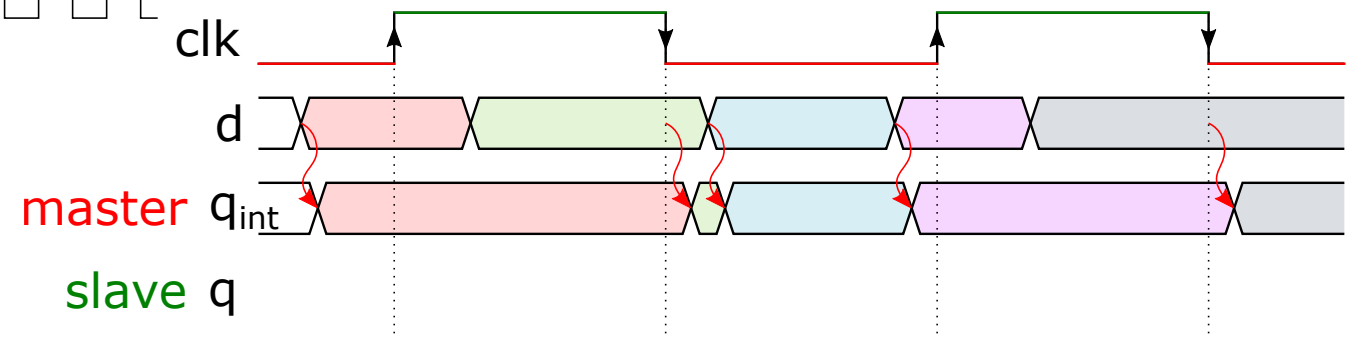
FlipFlops

D-FlipFlop

Master-Slave Latch



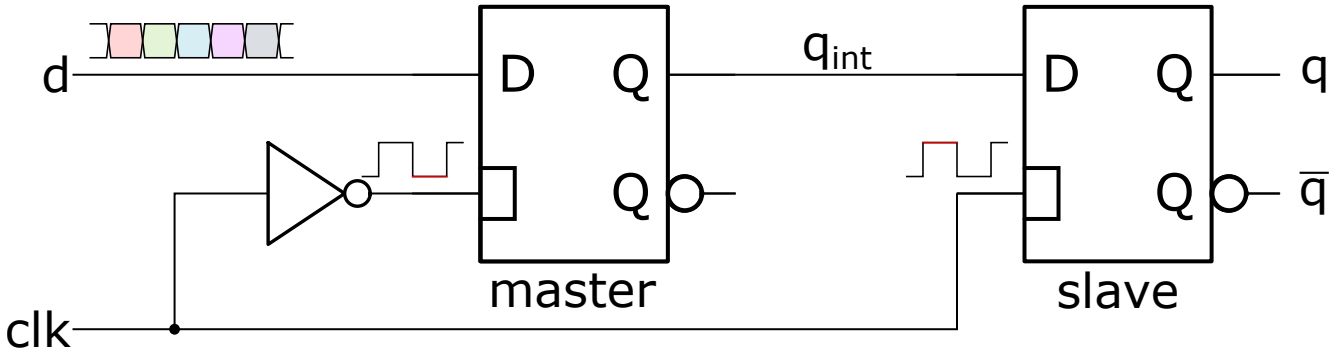


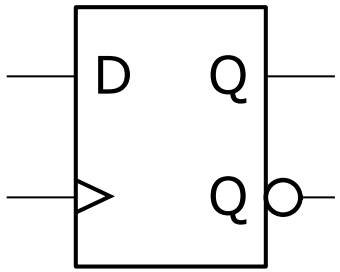


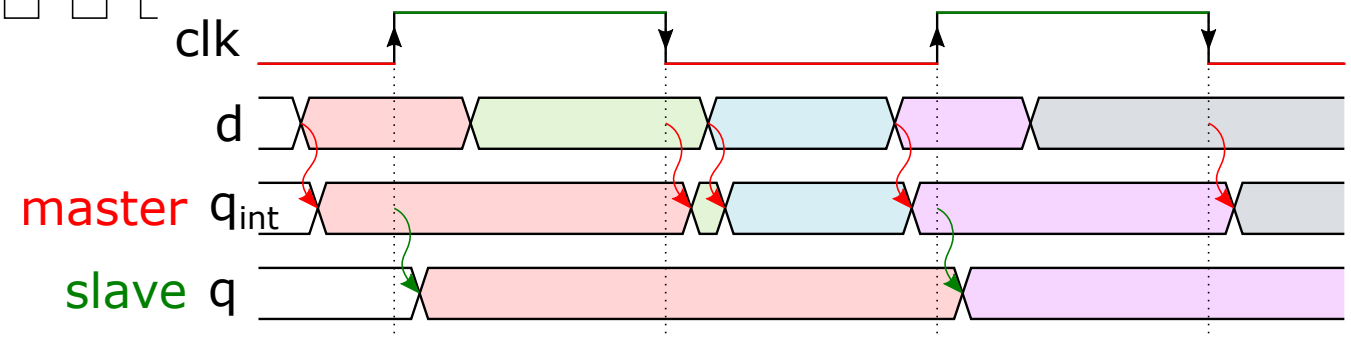
FlipFlops

D-FlipFlop

Master-Slave Latch



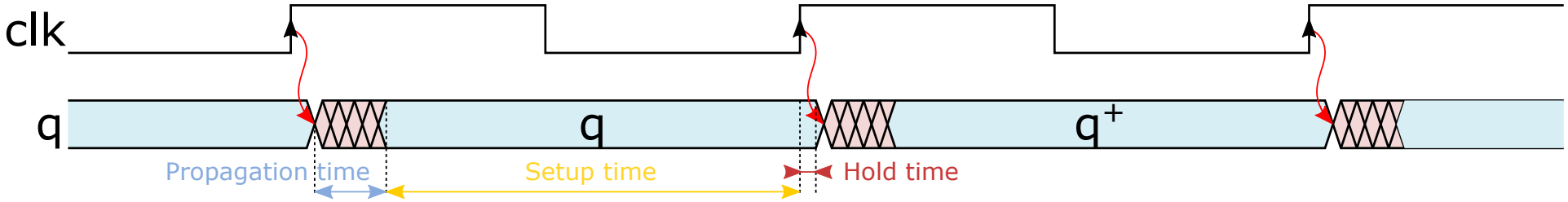




FlipFlops

D-FlipFlop

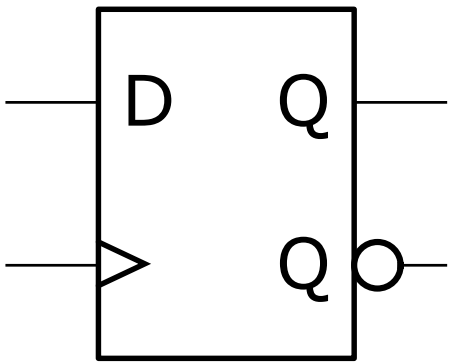
- Alle Signale sind vor der steigenden Flanke des Taktsignales stabil (Einhaltung von Setup Time und Hold Time).
- Nach dieser steigenden Flanke ändern sich alle Flip-Flop-Ausgänge mit unterschiedlichen Verzögerungen (Laufzeit).
- Der nächste Wert, den der Ausgang q bei der nächsten Taktflanke annimmt, wird mit "q+" angegeben (gegeben durch die Gleichung und die Schaltung, die sich am Eingang des Flipflops befindet).



D-FlipFlop

Charakteristische Gleichung

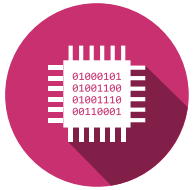
D	Q	Q^+	Funktion
0	0	0	Laden des D-Wertes bei der steigenden Flanke des Taktsignals
0	1	0	
1	0	1	
1	1	1	



$$q^+ = d$$

Aufgabe 3.1

Aufspüren von Übergängen

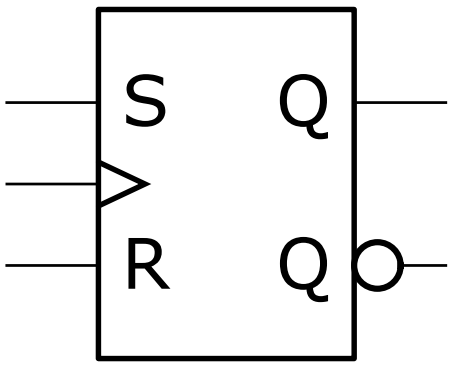


Mit Hilfe von eines D-Flipflop‘s und von logischen Gattern, entwerfen Sie eine Schaltung, welche die Übergänge ihres Eingangssignal aufspürt.

FlipFlops
SR-FlipFlop

Heutzutage ersetzt durch D-FlipFlops

S	R	Q^+	Funktion
0	0	Q	Speicherung
0	1	0	reset
1	0	1	set
1	1	1	-



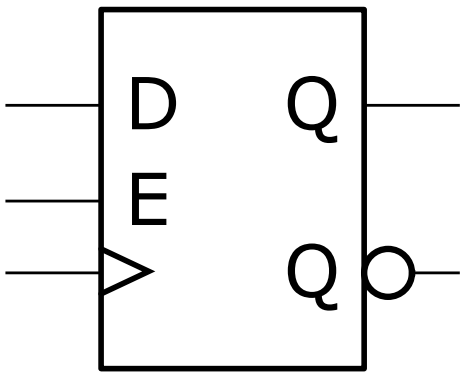
$$q^+ = s + \overline{r}q$$

FlipFlops

E-FlipFlop

Erlaubt es mit verschiedenen Geschwindigkeiten zu arbeiten

E	D	Q^+	Funktion
0	0	Q	Speicherung
0	1		
1	0	0	Laden des Wertes D (Abtastung)
1	1	1	

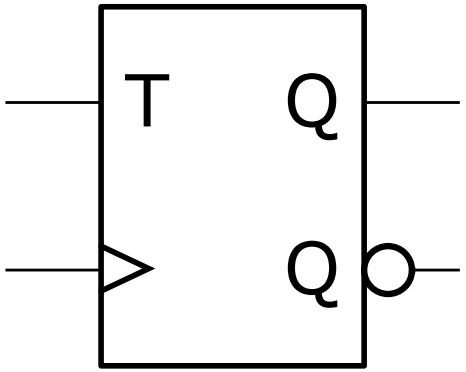


$$q^+ = ed + \overline{e}q$$

FlipFlops
T-FlipFlop

Erlaubt es Zählerschaltungen zu vereinfachen

T	Q^+	Funktion
0	Q	Speicherung
1	$\bar{Q}$	Invertierung



$$q^+ = t \oplus q$$

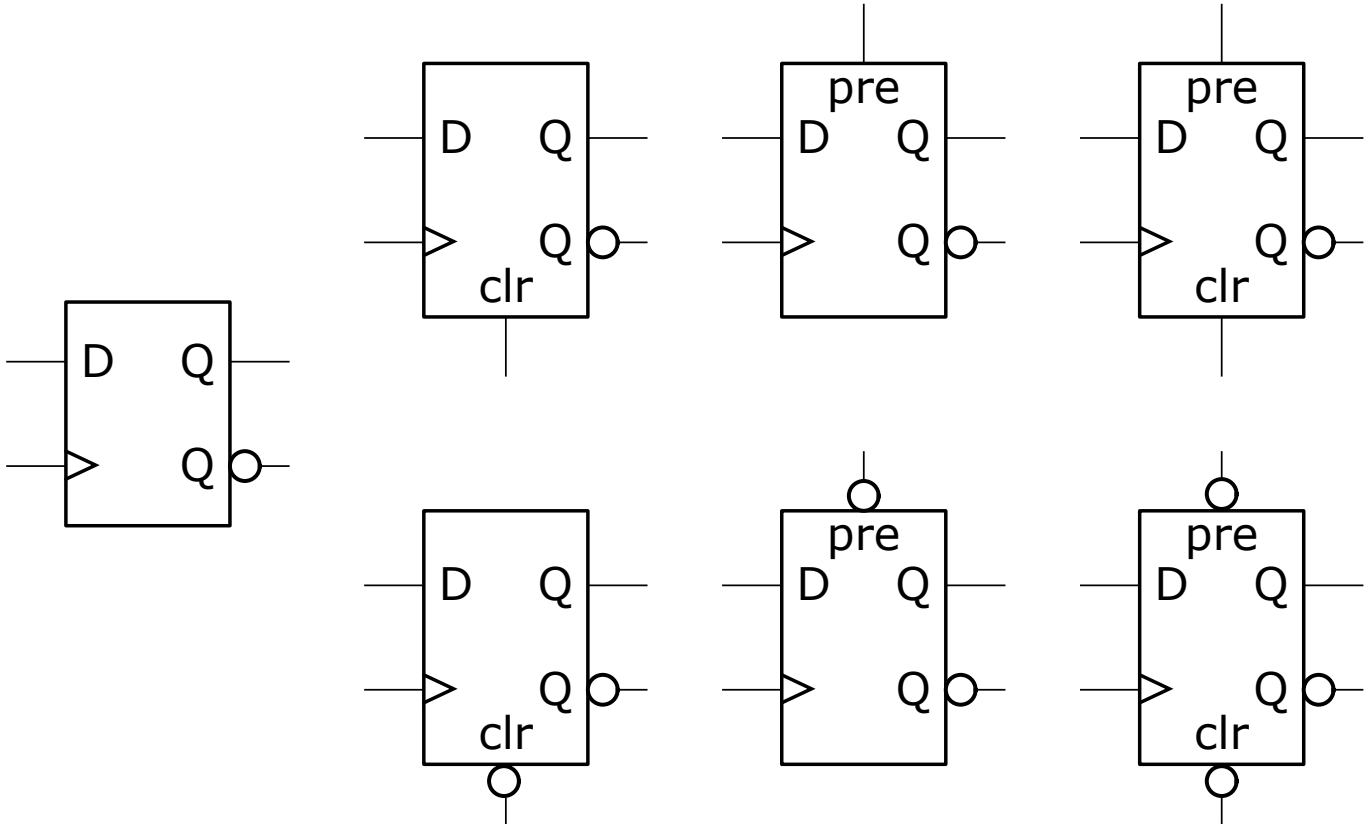
Aufgabe 3.6

Schieberegister

Mit Hilfe von T-Flipflops, erstellen Sie einen 4-Bit Schieberegister.

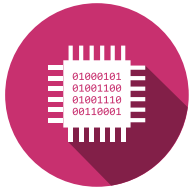
FlipFlops

Asynchrone Eingänge



Aufgabe 3.7

Asynchrone Nullsetzung



Mit Hilfe von einem RC-Glied und von logischen Gattern, erstellen Sie einen Schaltkreis zur Initialisierung der Flipflops beim Einschalten der Elektronik.

Referenzen

- [Wak00] (Englisch) Sehr vollständige Presentation
- [Kün97] (Deutsch) Gute Presentation
- [Max95] (Französisch) Gute Presentation

QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE SQUIRRELS

WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK

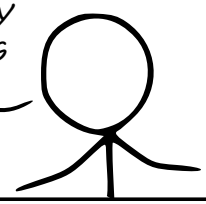
WHY ARE THERE DINOSAUR GHOSTS
WHY AREN'T THERE
WHY IS THERE HELL IF

WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY DO IGUANAS DIE

WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE SWISS AFRAID OF DRAGONS

WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTER
WHY IS HTTPS IMPORTANT

WHY AREN'T MY ARMS GROWING



WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY IS YKK ON ALL ZIPPERS

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM
WHY IS THERE LAVA
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE DOGS AFRAID OF FIR
WHY IS THERE NO KING IN EN

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE FEMALE

WHY ARE THERE GHOSTS

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE
WHY IS LIFE SO

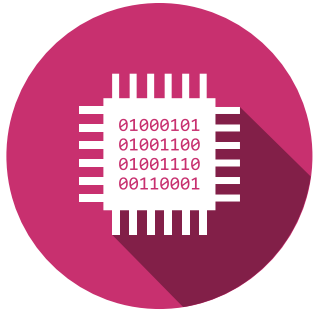
WHAT IS https://xkcd.com/1256/

WHY DO THEY SAY T-MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET

WHY AREN'T THERE GUNS IN

Hes·so VALAIS WALLIS

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



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